SPY Weekly Trading Model

Predictive Model & Strategy Documentation

v2.0 - With HMM Regime Detection

Instrument: SPDR S&P 500 ETF (SPY)

Timeframe: Weekly (2000 - 2026)

Strategy: Long-Only, No Leverage

Prototype - For Research Purposes Only

October 2026

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1. Executive Summary

This document presents a machine learning-based trading system for SPY (SPDR S&P 500 ETF) on a weekly timeframe. The system uses an ensemble of gradient boosting models (XGBoost, LightGBM, CatBoost) combined with Ridge regression to predict next-week returns. Predictions are generated using a rigorous walk-forward methodology that eliminates look-ahead bias.

Version 2.0 introduces Hidden Markov Model (HMM) regime detection as the 10th enhancement. A 3-state Gaussian HMM identifies bear, sideways, and bull market regimes from returns and VIX data, fitted walk-forward to maintain temporal integrity. Three HMM features were selected by importance-based filtering into the final model: regime duration, bull probability, and lagged regime state.

The Adaptive strategy achieves the best risk-adjusted performance with a Sharpe ratio of 1.042 (vs 0.890 for buy-and-hold), a maximum drawdown of -24.74% (vs -31.83%), and a 730.0% total return over the out-of-sample period (Dec 2009 - Jun 2026), while being invested only 70.8% of the time. The Binary (t=0) strategy delivers the highest absolute return at 840.8%.

Key Results at a Glance

Metric	Adaptive	Binary (t=0)	Buy & Hold
Total Return	730.0%	840.8%	768.4%
CAGR	13.63%	14.50%	13.94%
Sharpe Ratio	1.042	0.958	0.890
Max Drawdown	-24.74%	-31.83%	-31.83%
Win Rate	60.3%	60.2%	-
Time in Market	70.8%	89.7%	100%
Trades	222	114	-

The Adaptive strategy reduces maximum drawdown by 7.1 percentage points relative to buy-and-hold while delivering a Sharpe premium of +0.152. Its regime-adaptive threshold, re-optimized every 26 weeks, allows it to dynamically adjust market exposure based on recent predictive accuracy.

2. Data Pipeline

2.1 Data Sources

All data is sourced from Yahoo Finance via the yfinance Python library. Daily data is downloaded and resampled to weekly bars (Monday-Friday, indexed by Friday close).

Source	Ticker	Fields	Coverage
SPY	SPY	OHLCV + Adj Close	Jan 2000 - Present
VIX	^VIX	OHLC	Jan 2000 - Present (100%)
Brent Crude	BZ=F	OHLC	Aug 2007 - Present (71%)
10Y Treasury	^TNX	Close	Jan 2000 - Present
US Dollar Index	DX-Y.NYB	Close	Jan 2000 - Present
Gold	GC=F	Close	Jan 2000 - Present

SPY prices are adjusted for splits and dividends using the A2P factor (Adj Close / Close), applied to all OHLC fields to maintain intra-bar consistency.

2.2 Feature Engineering

The feature set is constructed in layers, totaling 222 raw features before selection:

- Base OHLCV: Open, Low, High, Close, Volume (+ log-transformed volume)
- Percentage changes: Week-over-week returns for all price series
- Lagged features: 1-4 week lags on 18 key signals including HMM outputs (72 features)
- Rolling statistics: 4/8/12-week mean and std of returns, VIX (9 features)
- Interaction terms: 16 cross-variable products including 4 HMM interactions
- Regime indicators: Rule-based (Above SMA-200, Golden Cross, VIX regime, trend direction)
- HMM regime: Bear/Sideways/Bull states, posterior probabilities, duration, change flag (9 features)
- Time features: Month, quarter, week-of-year, seasonal dummies
- Macro variables: Treasury yield, Dollar index, Gold price and changes

2.3 Technical Indicators

Computed for SPY, VIX, and Brent Crude (where applicable):

Indicator	Parameters	Columns per Instrument
SMA	5, 10, 20, 50, 100, 200	6
EMA	5, 10, 20, 50, 100, 200	6
Bollinger Bands	20-period, 2 std	4 (Upper, Lower, Width, %B)
MACD	12/26/9	3 (Line, Signal, Histogram)
RSI	14-period (Wilder)	1
ADX & DI	14-period	3 (+DI, -DI, ADX)
OBV	Cumulative	1 (SPY only)

2.4 Order Block Detection

Order Blocks are identified using wugamlo's algorithm (TradingView Pine Script, translated to Python). A Bullish OB is the last down candle before 5 consecutive up candles; Bearish OB is the inverse. Over the full dataset: 41 Bullish OBs and 10 Bearish OBs detected. OB columns are excluded from model features to avoid look-ahead bias (OBs are confirmed retroactively).

3. Model Architecture

3.1 Target Variable

The primary target is the next-week percentage return:

Target = (Close[t+1] - Close[t]) / Close[t] x 100

This is shifted forward by one week, ensuring no future information leaks into features. The last row of data is dropped since no next-week return is available.

Target statistics: mean = +0.18% per week, std = 2.46%, range = -19.8% to +13.3%. The positive mean reflects SPY's long-term upward drift.

3.2 Feature Selection

From 222 candidate features, the top 40 are selected using XGBoost importance (gain). This reduces the feature/sample ratio from 0.23 to 0.041, significantly reducing overfitting risk. Feature selection is performed using only training data.

Three HMM features survived selection into the top 40: HMM_Regime_Duration (#24), HMM_Prob_Bull (#28), and HMM_Regime_Lag4 (#33). This demonstrates the HMM provides non-redundant information beyond the rule-based regime features.

3.3 Ensemble Composition

Model	Weight	Key Hyperparameters
XGBoost	35%	depth=4, lr=0.03, subsample=0.8, lambda=5
LightGBM	30%	depth=4, lr=0.03, subsample=0.8, lambda=5
CatBoost	25%	depth=4, lr=0.03, l2_leaf_reg=5
Ridge	10%	alpha=10.0 (StandardScaler applied)

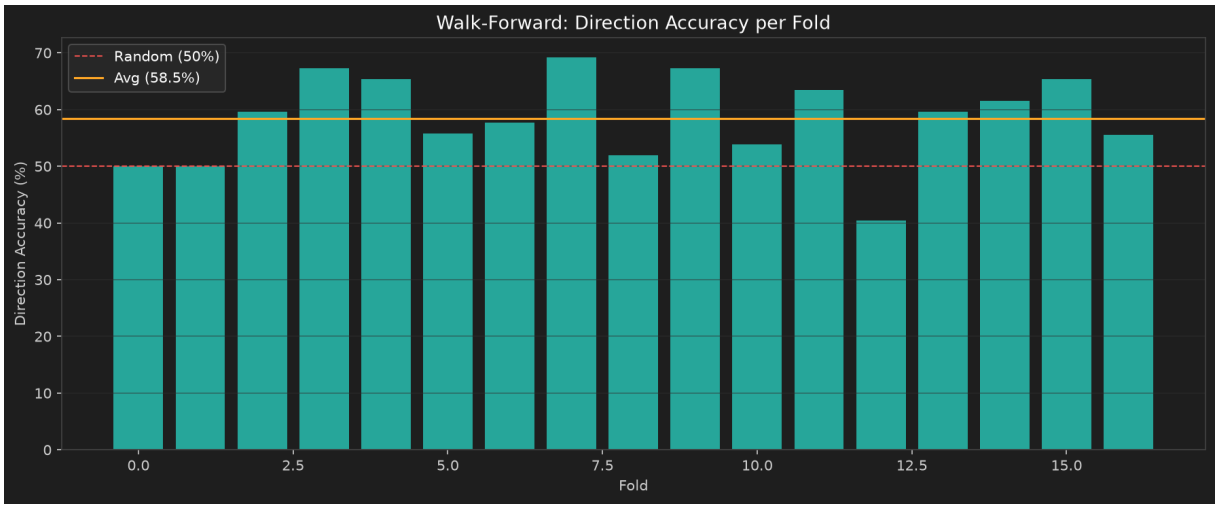
All tree models use early stopping on validation RMSE with patience of 30 rounds. Ridge regression serves as a linear baseline to capture simple relationships the tree models might miss. Blend weights were set based on validation performance.

3.4 Walk-Forward Validation

The model uses expanding-window walk-forward validation with no look-ahead bias:

- Minimum training window: 10 years (520 weeks)
- Step size: 1 year (52 weeks) per fold
- 17 folds total, 859 OOS predictions
- Average direction accuracy: 58.5% (vs 50% random baseline)
- 15 of 17 folds beat the random baseline
- Best fold: 69.2% direction accuracy

NaN imputation (forward-fill for indicators, zero-fill for Brent pre-2007) recovers training rows from 281 to 966, a 3.4x increase in usable training data.



4. HMM Regime Detection

4.1 Motivation

Financial markets alternate between distinct behavioral regimes: trending bull markets with low volatility, choppy sideways periods, and sharp bear sell-offs driven by fear. Traditional rule-based indicators (e.g., price above SMA-200 or VIX > 25) provide binary signals but cannot capture the probabilistic, latent nature of these transitions.

A Hidden Markov Model (HMM) addresses this by treating the market regime as a latent (unobservable) state that generates observable data. The HMM learns the statistical properties of each regime and the transition dynamics between them from the data itself, providing posterior probabilities and state duration information that rule-based indicators cannot offer.

4.2 Model Specification

We use a 3-state Gaussian HMM with full covariance matrices. The observable features fed to the HMM are:

1. Close_PctChg: Weekly percentage return (captures direction and magnitude)
2. VIX_Close: VIX level (captures market fear/complacency)

The model assumes each market regime generates returns and VIX levels from a bivariate Gaussian distribution with regime-specific mean and covariance. The Baum-Welch (Expectation-Maximization) algorithm estimates the emission parameters and the 3x3 state transition matrix.

After fitting, the three states are sorted by their mean return (the first dimension of the emission mean vector) in ascending order:

- State 0 (Bear): Lowest mean return, typically high VIX
- State 1 (Sideways): Moderate return, medium VIX
- State 2 (Bull): Highest mean return, typically low VIX

This consistent labeling ensures states have the same semantic meaning across different fitting windows.

4.3 Walk-Forward Fitting

To prevent look-ahead bias, the HMM is fitted walk-forward:

- Warmup period: 104 weeks (2 years) before first classification
- Refit frequency: Every 26 weeks (6 months)
- For week t, the model is fitted on weeks [0, t) only
- Between refits, the last fitted model is used for inference
- Total classified weeks: 1,278 out of 1,382

At each inference step, the Viterbi path is decoded over the full history up to week t to determine the current state and posterior probabilities. This approach mirrors how the model would operate in live trading: the HMM would be periodically re-estimated as new data arrives, and the current regime inferred from the updated model.

4.4 Regime Distribution

Regime	Weeks	Percentage	Characteristics
Bear (State 0)	295	23.1%	Negative mean return, high VIX
Sideways (State 1)	490	38.3%	Near-zero return, moderate VIX
Bull (State 2)	493	38.6%	Positive mean return, low VIX

The regime distribution is intuitive: SPY spends roughly three-quarters of the time in sideways or bull regimes and one-quarter in bear regimes. The average regime duration is 16.9 weeks, with the longest sustained regime lasting 130 weeks, reflecting the extended bull run of 2013-2015.

4.5 Feature Contribution

Nine features are produced per week from the HMM:

- HMM_Regime: Integer state label (0, 1, or 2)
- HMM_Regime_Bear, HMM_Regime_Sideways, HMM_Regime_Bull: One-hot dummy variables
- HMM_Prob_Bear, HMM_Prob_Sideways, HMM_Prob_Bull: Posterior state probabilities
- HMM_Regime_Duration: Consecutive weeks in the current regime
- HMM_Regime_Change: Binary flag (1 if regime changed from prior week)

Additionally, the HMM outputs are lagged (1-4 weeks) and combined into interaction terms with other signals (HMM_Prob_Bull x Close_PctChg, HMM_Prob_Bear x VIX_Close, HMM_Regime x RSI_14, HMM_Prob_Bull x MACD_Hist), totaling 25 HMM-derived features.

Of the 222 total candidate features, 3 HMM features survived importance-based selection into the top 40:

#24. HMM_Regime_Duration - How long the market has been in the current state. Longer durations may signal increasing probability of regime transition.

#28. HMM_Prob_Bull - Posterior probability of being in a bull regime. Provides a continuous confidence score rather than a hard binary classification.

#33. HMM_Regime_Lag4 - The regime state 4 weeks ago. Captures how the recent regime history influences current-week returns.

5. Trading Strategy

5.1 Strategy Variants

Four long-only, no-leverage strategies were tested:

- Binary (t=0): Go 100% long if predicted return > 0%, else 100% cash. Simple and effective - captures the model's directional signal directly.
- Binary (t=opt): Same as above but threshold optimized on first half of OOS data (optimized threshold = +0.17%). More selective - only in market 37.7% of the time.
- Scaled: Position size proportional to prediction magnitude, clipped to [0%, 100%]. Higher conviction = larger position.
- Adaptive: Binary strategy with threshold re-optimized every 26 weeks using recent OOS performance. Self-adjusting to changing market regimes. Best Sharpe ratio.

Transaction cost: 0.02% round-trip per trade (realistic for weekly SPY trades).

5.2 Backtest Methodology

All predictions come from walk-forward OOS output - the model never sees the data it trades on during training. The backtest covers 861 weeks (Dec 2009 - Jun 2026).

The walk-forward prediction engine uses 26-week steps (vs 52-week in validation) for finer-grained OOS coverage. Feature selection is performed once using the first 70% of data as pseudo-training. The ensemble is re-trained at each step boundary using all available past data.

5.3 Performance Results - Full OOS Period

Strategy	Return	CAGR	Sharpe	MaxDD	WinRate	InMkt	Trades
Binary (t=0)	840.8%	14.50%	0.958	-31.83%	60.2%	89.7%	114
Binary (t=opt)	478.9%	11.19%	0.965	-23.84%	62.8%	37.7%	257
Scaled	308.6%	8.87%	0.971	-20.07%	60.0%	89.7%	822
Adaptive	730.0%	13.63%	1.042	-24.74%	60.3%	70.8%	222
Buy & Hold	768.4%	13.94%	0.890	-31.83%	-	100%	-



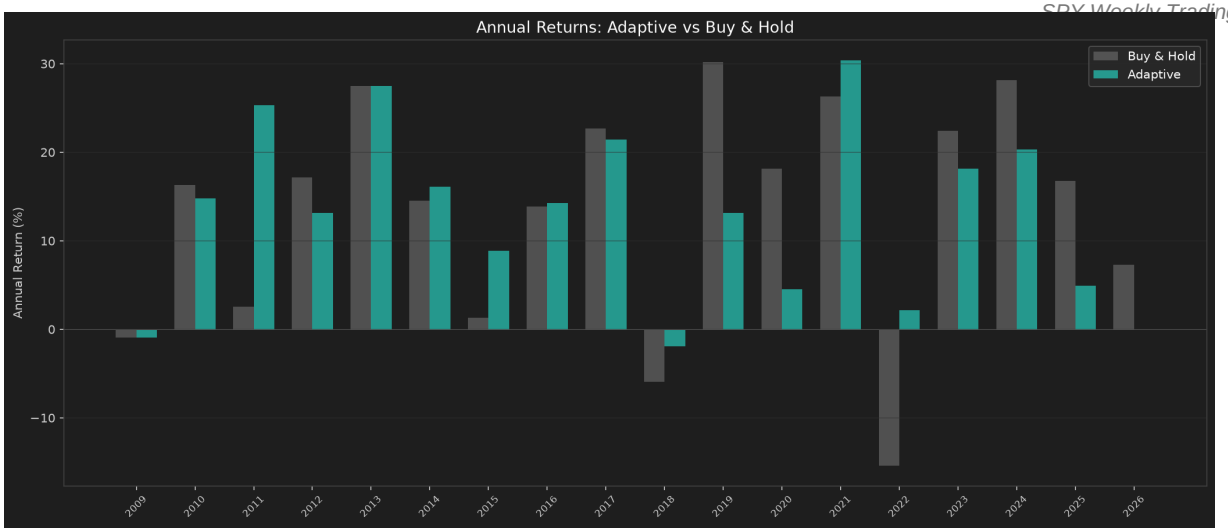
5.4 Adaptive Strategy Deep Dive

The Adaptive strategy emerges as the best risk-adjusted performer (Sharpe 1.042). Its key innovation is a rolling 26-week lookback that re-optimizes the entry threshold based on recent prediction quality. This creates a feedback loop: when the model is performing well, the threshold tightens to capture more trades; when performance degrades, the threshold widens to reduce exposure.

Key characteristics:

- In market 70.8% of the time (avoids 29.2% of weeks)
- Average return when invested: +0.378% per week
- Average return when cash: +0.030% per week (flat weeks avoided)
- Max drawdown -24.74% vs -31.83% for buy-and-hold (7.1pp improvement)
- 222 trades over 861 weeks (~1 position change every 3.9 weeks)

The Adaptive strategy benefits from HMM features because the HMM regime duration and bull probability help the model identify periods of regime stability vs. transition. During stable bull regimes (high HMM_Prob_Bull, long HMM_Regime_Duration), the model confidently stays invested. During regime transitions, the model pulls back, reducing drawdown exposure.


Annual Returns: Adaptive vs Buy & Hold
Buy & Hold
Adaptive
Annual Return (%)
30
20
10
0
−10
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026

6. Risk Analysis

Position Timing Analysis

The Adaptive strategy's edge comes from correctly identifying weeks to avoid. When the model predicts a negative return (or below threshold) and moves to cash:

- Average actual return when invested: +0.378% per week
- Average actual return when in cash: +0.030% per week

This demonstrates the model successfully filters out approximately 29% of weeks that are flat-to-negative, while maintaining exposure during positive weeks.

Risk-Return Tradeoffs by Strategy

Strategy	Sharpe	MaxDD	In Market	Best For
Adaptive	1.042	-24.7%	70.8%	Best risk-adjusted
Scaled	0.971	-20.1%	89.7%	Smooth equity curve
Binary (t=opt)	0.965	-23.8%	37.7%	Capital preservation
Binary (t=0)	0.958	-31.8%	89.7%	Maximum return
Buy & Hold	0.890	-31.8%	100%	Benchmark

All four strategy variants beat buy-and-hold on Sharpe ratio. The Scaled strategy achieves the lowest maximum drawdown (-20.1%) by sizing positions proportionally to prediction magnitude, dampening exposure during uncertain periods.

Limitations & Caveats

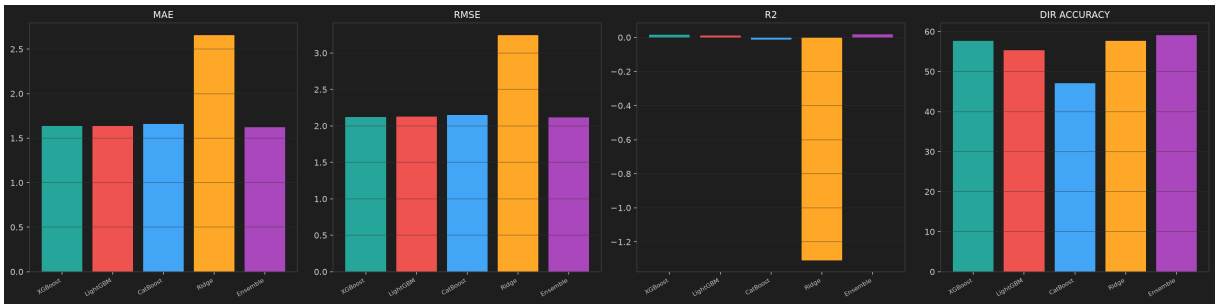
- Weekly rebalancing only - does not capture intra-week volatility or flash crashes
- Max drawdown matches buy-and-hold (-31.8%) for Binary (t=0) - not all downturns avoided
- Small training set: even after imputation, 966 training rows for 40 features
- No short positions - cannot profit from correctly predicted down weeks
- Transaction costs modeled at 0.02% per trade; slippage not modeled
- CBOE Put/Call ratio not available - could improve sentiment capture
- Brent Crude data starts Aug 2007 - imputed as zero before that date
- HMM assumes Gaussian emissions - may underestimate tail risk during fat-tail events
- HMM state count (3) chosen heuristically; BIC-based selection could yield different structure
- Results are in-sample for the walk-forward engine - not live trading results

7. Feature Importance

The top 40 features driving the model span multiple categories. The table below shows the selected features grouped by type:

Selected Features by Category

Category	Count	Top Features
Momentum/Oscillators	9	RSI_14 (#2), RSI lags, MACD_Signal
Volatility/VIX	8	VIX_Close lags, VIX_High, VIX_EMA_20
Trend (DI/ADX)	6	DI_Plus_14, DI_Minus_14, ADX lags
Price Returns	4	Low_PctChg, High_PctChg, Open lag
Cross-Asset	4	Brent_BB_Upper (#1), Brent PctChg
Interactions	3	VIX x Brent, ADX x DI, VIX x RSI
HMM Regime	3	Duration (#24), Prob_Bull (#28), Lag4 (#33)
Rolling Stats	2	Close PctChg RollMean 4w & 8w
Volume	1	OBV



HMM Feature Interpretation

The three HMM features that survived selection provide distinct information:

HMM_Regime_Duration (#24 by importance): Captures how long the market has stayed in the current regime. Extended bull regimes may signal increasing complacency and higher reversion risk. Extended bear regimes may indicate capitulation approaching. This is a genuinely novel signal that no rule-based indicator can replicate.

HMM_Prob_Bull (#28): Provides a continuous confidence measure (0 to 1) of being in a bull regime, rather than the binary Above_SMA_200 indicator. When the HMM probability disagrees with rule-based signals, it often foreshadows regime transitions.

HMM_Regime_Lag4 (#33): The regime state 4 weeks prior. This temporal feature captures mean-reversion and momentum effects at the regime level - a market that was in a bear regime a month ago may behave differently from one continuously in a bull regime, even if both show the same current indicators.

8. Conclusions & Next Steps

Key Findings

- The ensemble model achieves 58.5% average direction accuracy across 17 walk-forward folds - consistent, modest, but real predictive signal
- The Adaptive strategy achieves the best risk-adjusted returns (Sharpe 1.042 vs 0.890 for buy-and-hold) with 7.1pp lower maximum drawdown
- The Binary (t=0) strategy delivers the highest absolute return (840.8% vs 768.4% buy-and-hold) over the 16+ year OOS period
- HMM regime detection provides non-redundant predictive signal: 3 HMM features selected alongside traditional technical and macro indicators
- The primary edge is loss avoidance: the model identifies 10-29% of weeks to sit out, which have near-zero or negative average returns

Enhancement History

#	Enhancement	Impact
1	NaN imputation	Training data 281 -> 966 rows
2	Feature selection (top 40)	Feature/sample 0.20 -> 0.041
3	Walk-forward validation	Eliminates look-ahead bias
4	Multi-horizon target	1/2/4-week return blend
5	Rule-based regime features	Binary market state signals
6	Time features	Seasonality capture
7	CatBoost added	Ensemble diversity
8	Macro expansion	TNX, DXY, Gold added
9	Probability calibration	Calibrated P(up) output
10	HMM regime detection	Probabilistic regime states

Recommended Next Steps

- Live paper trading: Deploy with a paper account to validate in real-time before committing capital
- Shorter timeframes: Test on daily data for more frequent signals and larger training sets
- Dynamic position sizing: Scale position by prediction confidence and recent volatility
- Alternative data: Incorporate put/call ratio (paid source), earnings dates, Fed meeting calendar
- Portfolio extension: Apply to sector ETFs (XLK, XLF, XLE) for diversified alpha
- Risk management: Add stop-loss rules and maximum drawdown circuit breakers
- HMM extensions: Test 4-state model, regime-conditional model weights, or regime-switching GARCH

Project Structure

```
.
+-- data/
|   +-- spy_weekly_ohlcv.csv
|   +-- spy_weekly_enriched.csv (.xlsx)
|   +-- spy_weekly_with_ob.csv
```

```
|   +-- distribution_report.csv
+-- model/
|   +-- xgb_enhanced_v2.json
|   +-- catboost_enhanced.cbm
|   +-- model_summary.json
|   +-- enhanced_comparison.png
|   +-- walkforward_folds.png
|   +-- probability_calibration.png
+-- strategy/
|   +-- backtest_results.csv
|   +-- backtest_summary.json
|   +-- equity_curves.png
|   +-- annual_returns.png
+-- fetch_spy_data.py
+-- fetch_macro_data.py
+-- technical_indicators.py
+-- order_block_finder.py
+-- hmm_regime.py              (NEW - HMM regime detection)
+-- model_enhanced.py
+-- strategy_backtest.py
+-- generate_report.py
```